

have developed systems that read brainwaves—in the form of electroencephalogram (EEG) signals—in order to help the disabled control wheelchairs, play games or type on a computer. Two companies are already preparing to sell such devices to mainstream consumers.

Australian company Emotiv will release a helmet equipped with 16 sensors that make it possible to direct twelve different movements in a computer game. The helmet can also detect emotions, says the company. It will be compatible with any PC running Windows and ship later this year for \$299.

Californian company NeuroSky too has come up with a device that can detect emotions: the firm says it can tell whether you are focused, relaxed, afraid or anxious, for example. NeuroSky does not sell it directly to the public. Instead, it is licensing its technology to other companies including Mattel, Nokia and Sega. These licensed developers will bundle the technology with their applications and sell into the retail market. Mattel, for example, will soon sell a game that involves players levitating a ball using thought alone.

## Wearable electronics

Fashion designers are integrating gadgets like iPods and GPS into the fabrics rather than merely providing storage for them. These designs may look like costumes borrowed from a sci-fi thriller. For instance, a dress can sense the mood of its wearer by gauging his gestures and then play an appropriate song from its MP3-integrated hood.

O'Neill, an outerwear company, has come up with a garment line that includes the GPS-imbudded NavJacket, which features LED arrow indicators on the sleeves that direct a skier to various routes and slopes.

Erik De Nijs, a student at the Utrecht School of the Arts in the Netherlands, has created a pair of tech jeans that incorporates a wireless Bluetooth keyboard into the lap of the pants, speakers into the knees of the jeans and a mouse in the back pocket. Other garments use a nickel-titanium shape-



O'Neill GPS-imbudded NavJacket

memory alloy to move shimmering panels of fabric as if they are breathing, like coral shifting with the tide.

## Simulated brain closer to thought

Researchers have constructed a detailed simulation of a small region of a brain molecule by molecule and recreated experimental results from real brains. When put in a virtual body, it gives the first indications of the molecular and neural basis of thought and memory.

While many computer simulations have attempted to code in 'brain-like' computation or to mimic parts of the nervous systems and brains of a variety of animals, the Blue Brain project was conceived to reverse-engineer mammal brains from real laboratory data and build up a computer model down to the level of the molecules that make them up.

The first phase of the project, which is now complete, modeled the neocortical column—a unit of the mammalian brain known as the neocortex which is responsible for higher brain functions and thought. The neocortical column can be thought of as an isolated proces-

sor. It is being integrated into a virtual reality agent—a simulated animal in a simulated environment—so the researchers will be able to observe the detailed activities in the column as the animal moves around the space.

"It starts to learn and remember things. We can actually see when it retrieves a memory, and where they retrieved it from because we can trace back every activity of every molecule, every cell, every connection and see how the memory was formed," said Prof. Henry Markram, leader of the Blue Brain project and founder of the Brain Mind Institute in Switzerland.

The next phase of the project will make use of a more advanced version of the IBM Blue Gene supercomputer that was used in the research to date. Such an approach would allow researchers to simulate how individuals will respond to a given drug or treatment.

## Facial barcodes help identify people

British scientists have shown that nearly all the information that is needed to identify human faces is contained in horizontal lines, such as those of eyebrows, eyes and lips. They arrived at this finding by manipulating photos of the faces of George Clooney, Coldplay singer Chris Martin and other celebrities.

These features could be simplified into B&W lines of information, i.e., barcodes. So these natural barcodes of information, built into the human face, could have significant implications for facial-recognition software.

## 'Instant on' computers

Ultra-thin ferro-electric materials used in smart cards will soon make computers more efficient by enabling them to switch on instantly.

Smart cards rely on ferro-electric materials to instantly reveal and update stored information; for example, in ATMs. A computer with this capability could instantly provide information and other data to the user.

Researchers led by Darrell Schlom